

INVESTIGATION REPORT

# Issue 1019: Ready handshake in the divide and square-root unit

openhwgroup/cv32e40p

|                       |                                                         |
|-----------------------|---------------------------------------------------------|
| <b>Document</b>       | RTL-TRIAGE-CV32E40P-1019                                |
| <b>Disposition</b>    | Dead code, proven                                       |
| <b>Record date</b>    | 2026-09-17                                              |
| <b>Classification</b> | Internal working document. Not a verification sign-off. |

## 1 Status

|                      |                                                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Issue</b>         | openhwgroup/cv32e40p#1019                                                                                           |
| <b>Issue filed</b>   | 2024-07-02                                                                                                          |
| <b>Issue type</b>    | Coverage hole referred for disposition                                                                              |
| <b>Affects</b>       | cv32e40p                                                                                                            |
| <b>Configuration</b> | RTL 6033d2b, COREV_PULP=1 FPU=1 (the upstream PULP_F0 configuration), top cv32e40p_formal_top, yosys-slang frontend |
| <b>Disposition</b>   | Dead code, proven                                                                                                   |
| <b>Severity</b>      | Not assessed                                                                                                        |
| <b>Agreement</b>     | Same disposition as the independent human record.                                                                   |
| <b>Evidence</b>      | 4 cited, 4 present                                                                                                  |

*Disposition meaning. The code cannot be reached in any state permitted by the stated model and assumptions, established by an unbounded proof.*

*Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.*

## 2 Description

The uncovered branch at fpnew\_divsqrt\_th\_32.sv:287 and 288 is unreachable. Line 286 opens the case where a new operation starts while the unit is ready; line 287 asks whether a write-back is completing in the same cycle, and line 288 keeps the unit ready if so. The unit cannot both accept a new operation and retire an old one in the same cycle, so line 288 is never executed. An unbounded proof (abc pdr, top cv32e40p\_formal\_top, COREV\_PULP=1 FPU=1) passes with the checker bound into the fpnew\_divsqrt\_th\_32 instance inside the CV32E40P FPU wrapper, not standalone. A bounded cover run reaches the line 286 antecedent at step 12, so the assertion is not vacuous. This is the same property as the upstream assert\_unreachable\_divsqrt\_th\_288, checked with a different tool chain.

## 3 Conditions

1. legal OBI handshakes on both the instruction and the data interface
2. scan\_cg\_en\_i=0
3. no CSR write to the hardware-loop CSRs (0xCC0, 0xCC1, 0xCC2, 0xCC4, 0xCC5, 0xCC6)
4. clock gates modelled as transparent
5. every trace starts in reset
6. instruction memory contents otherwise unconstrained

*Reproduced from the assumes that are enabled in the upstream formal environment at cores/cv32e40p/scripts/formal/src, plus the two modelling choices the open-source flow requires.*

## 4 Implications

*Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.*

## 5 Action

### Waiver text, to be applied to the coverage exclusion

fpnew\_divsqrth\_32.sv:287-288 (unit\_ready\_d held high when a write-back completes in the same cycle a new operation starts): formally proven unreachable (abc pdr, unbounded, bound into the fpnew\_divsqrth\_32 instance inside cv32e40p\_formal\_top, COREV\_PULP=1 FPU=1, RTL 6033d2b). op\_starting with unit\_ready\_q and ex2\_inst\_wb with ex2\_inst\_wb\_vld\_q are mutually exclusive. Assumptions: legal OBI handshakes on both interfaces, scan\_cg\_en\_i=0, no CSR write to the hardware-loop CSRs, transparent clock gates. Evidence: runs/v1-cv32e40p-1019/formal/p\_assert/sby.log (PASS) and formal/c\_antecedent/sby.log (antecedent reached at step 12).

### Checker used

checker\_1019.sv

## 6 Evidence

| Ref | Type  | Path, relative to the run directory | File    | Result recorded                                                                                                                                       |
|-----|-------|-------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1  | proof | formal/p_assert/sby.log             | present | abc pdr PASS (COREV_PULP=1 FPU=1): op_starting with unit_ready_q never coincides with ex2_inst_wb and ex2_inst_wb_vld_q                               |
| E2  | proof | formal/p_assert/result.json         | present | machine-readable status record for the proof job                                                                                                      |
| E3  | proof | formal/c_antecedent/sby.log         | present | smtbmc/yices cover PASS (depth 80): op_starting with unit_ready_q is reached at step 12 inside the real FPU instance, so a_divsqrth288 is not vacuous |
| E4  | proof | formal/c_antecedent/result.json     | present | machine-readable status record for the cover job                                                                                                      |

*Paths are relative to runs/v1-cv32e40p-1019. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.*

### What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.

## 7 Limitations and scope

- The proof establishes unreachable under the model and the environment assumptions named in Status and Conditions. It does not establish that the surrounding logic is correct.
- This is an internal investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Public demonstration runs on open cores are not client engagements and not production verification results.

## 8 Provenance and measurement

|                            |                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Run directory</b>       | runs/v1-cv32e40p-1019                                                                                                                       |
| <b>Inputs shown</b>        | title, body(links stripped), labels                                                                                                         |
| <b>Human record</b>        | Dolphin Design 2024: waived in v2; reported proven unreachable with Siemens Questa Static formal, assertion assert_unreachable_divsqrth_288 |
| <b>Assessed confidence</b> | 0.9. This is the investigator's own estimate. It is not a measurement and not a                                                             |

|                   |                     |
|-------------------|---------------------|
|                   | reliability figure. |
| <b>Iterations</b> | 1                   |

*Generated by reporting/scripts/issue\_document.py from runs/v1-cv32e40p-1019/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.*